

BUILDER JOURNAL

Modern Object Detection Systems Comparison

FELLOW

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TRACK & ASSIGNMENT

Computer Vision · Assignment 6

PROGRAM

Research Report

SYSTEM OVERVIEW

Research Scope: A comprehensive empirical comparison analyzing four landmark architectural frameworks shaping the deep learning object detection field: YOLO, Faster R-CNN, SSD, and DETR. Each model represents distinct architectural philosophies spanning unique trade-offs over inference speeds, tracking bounds, and structural complexities.

"Giving machines the ability to see is not just a technical problem — it is the foundation of every system that must actively interact with and interpret the physical world."

01/ ARCHITECTURAL FRAMEWORK OVERVIEWS

Modern paradigms have evolved from legacy hand-crafted layouts into high-performing networks categorizing distinct bounding regressions:

01/ REGRESSION · YOLO (YOU ONLY LOOK ONCE)

Core Innovation: Framed object tracking as a single continuous regression challenge, executing grid-cell spatial bounding box generations and classification passes concurrently in one unified neural forward pass.

ARCHITECTURAL SPECS & TRADE-OFFS

Strengths: Fastest framework available; YOLOv8 Nano maps real-time tracking sequences at 160+ FPS on standard GPU configurations. Incorporates anchor-free heads, C2f gradient loops, and decoupled structural branches.

Weaknesses: Exhibits slight precision drop boundaries when processing highly congested or microscopic item occlusions.

02/ TWO-STAGE · FASTER R-CNN

Core Innovation: Implements a deep Two-Stage pipeline parsing region nominations through a dedicated Region Proposal Network (RPN) sharing downstream feature map parameters before processing ultimate classification regions.

ARCHITECTURAL SPECS & TRADE-OFFS

Strengths: Highest absolute evaluation precision mapping, specifically optimizing complex overlapping structures or fine-grained medical imaging diagnostics.

Weaknesses: Constrained throughput (5–15 FPS), requiring heavier GPU memory spaces, which makes real-time continuous video stream pipelines impossible.

03/ MULTI-SCALE · SSD (SINGLE SHOT MULTIBOX DETECTOR)

Core Innovation: Reconciles single-shot compute speedups with multi-scale feature maps, mapping predictions directly from multi-layered networks at variable high/low target visual resolutions.

ARCHITECTURAL SPECS & TRADE-OFFS

Strengths: High-speed tracking running seamlessly inside lower-spec resource limits like edge processors, mobile applications, or battery-constrained hardware setups.

Weaknesses: Lower global accuracy margins compared to both high-tier YOLO models and traditional multi-stage networks.

04/ TRANSFORMER · DETR (DETECTION TRANSFORMER)

Core Innovation: Removes custom components like anchor configurations and Non-Maximum Suppression (NMS) routines entirely by defining object detection as an elegant end-to-end set prediction problem using global Transformer Encoders/Decoders coupled with bipartite matching.

ARCHITECTURAL SPECS & TRADE-OFFS

Strengths: Exceptional mastery mapping comprehensive global contexts, scene relationships, and long-range structural dependencies natively without duplicates.

Weaknesses: Requires heavy compute bounds to converge during training phases (500+ epochs vs 50 epochs for YOLO structures).

02/ EMPIRICAL PERFORMANCE MATRIX

MODEL FRAMEWORK	INFERENCE SPEED	ACCURACY (MAP)	MEMORY ALLOCATION	OPTIMAL REAL-WORLD DEPLOYMENT MODE
YOLOv8	... 30–160+ FPS (Fastest)	... High (53.9 mAP)	... Low VRAM (~2 GB)	Real-time production, Edge devices, surveillance tracking
Faster R-CNN	... 5–15 FPS (Slowest)	... Very High (42.7 mAP)	... High VRAM (~6 GB)	Medical imaging, Satellite analysis, Offline research pipelines
SSD	... 25–60 FPS (Fast)	... Lower (25.1 mAP)	... Very Low VRAM (~1 GB)	Mobile applications, Embedded IoT networks, CPU units
DETR	... 10–28 FPS (Medium)	... High (42.0 mAP)	... Medium VRAM (~4 GB)	Complex multi-object scenes, Panoptic context mapping

03/ CORE ANALYTICAL DECISIONS

One-Stage vs Two-Stage Paradigms

One-stage designs (YOLO, SSD) handle detection via a single network traversal layer to prioritize execution velocity, whereas two-stage layouts (Faster R-CNN) intentionally run secondary validation cycles, swapping execution speeds for raw spatial boundary accuracy.

Strategic Framework Selection

For our fellowship's core ****Bottle Detection System****, YOLOv8 Nano was purposefully chosen. It successfully combines ultra-lightweight parameters (6.2MB size running entirely in memory) with strong detection results (87.1% mAP50 and 93.5% Precision) while running at a processing rate of 9.3ms per image on a Quadro T1000 GPU configuration.

The Critical Role of Transfer Learning

All four paradigms rely heavily on transfer learning structures. Starting training pipelines with pre-trained COCO dataset base weights allows models to inherit complex basic features (edges, lighting shapes, contours) instantly. This isolates subsequent training runs down to specialized final layers, cutting development overhead down by 90%+.

04/ STRATEGIC FUTURE ROADMAP

- **Real-Time Transformers (RT-DETR):** Merging the superior global structural matching accuracy of standard transformers with high-speed YOLO pipeline velocities.
- **Zero-Shot Foundation Architectures:** Testing open-vocabulary foundation blocks (e.g., SAM, Grounding DINO) to handle unseen object classes via textual prompting.
- **On-Device Optimization:** Refining compact weight configurations using CoreML and TFLite execution layouts to achieve high-performance edge inference without cloud server dependencies.